



NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2025-2026

Batch Number	AB-1
Team Members	G Revanth Sai -(22471A0521) M Karthik -(22471A0528) Ch Jaswanth Reddy -(22471A0506) K Adithya -(22471A0527)
Guide	Dr.Rama Krishna Eluri
Title	STCNN: Combining SMOTE-TOMEK With CNN for Imbalanced Classification of Alzheimer's Disease
Domain/Technology	Deep LEARNING
Base Paper Link	https://ieeexplore.ieee.org/document/10413497
Dataset Link	https://www.kaggle.com/datasets/marcopinamonti/alzheimer-mri-4-classes-dataset
Software Requirements	Browser: Any latest browser like Chrome Operating System: Windows 7 Server or later Python
Hardware Requirements	System Type: Intel Core i5 or above RAM: 8 GB Number of cores:5 Number of Threads: 4
Abstract	Alzheimer's Disease (AD), the leading cause of dementia, poses significant diagnostic challenges due to limitations in current methods like MRI analysis and cognitive testing. This paper introduces a deep learning framework, STCNN, which combines Convolutional Neural Networks (CNN) with SMOTE-TOMEK resampling to address class imbalance in MRI datasets. Using brain MRI images from Kaggle, the model accurately classifies different stages of dementia. The STCNN model achieves 99.36% accuracy and 99% F1-score, outperforming existing models in accuracy, efficiency, and performance.

Signature of the student(s)

Signature of the Guide

Signature of the project coordinator